Preliminary Exploration: ROS2 – ns-3 Wi-Fi 7 Integration Pipeline

Motivation

Modern robotic systems increasingly rely on wireless links for distributed sensing, control, and coordination. However, Wi-Fi performance—latency, jitter, throughput, and packet loss—can vary significantly under different network conditions and PHY/MAC configurations.

To better understand these effects, I began developing a software-in-the-loop (SIL) environment combining:

- real ROS2 nodes
- ns-3 Wi-Fi 7 (802.11be) simulations.

This work was part of my thesis preparation and is still in an exploratory stage..

1. Prototype Architecture Overview

```
ROS2 Talker i (container)
UDP Send → 172.17.0.X : (9000 + i)
   ↓
   ↓ real Linux UDP (port 9000 + i)
   ↓
ns-3 STA Node i
Bind: 0.0.0.0 : (9000 + i)
SendTo: AP : (5000 + i)
   ↓
   ↓ Wi-Fi 7 PHY/MAC simulation (802.11be)
   ↓
ns-3 AP Node
Bind: 0.0.0.0 : (5000 + i)
SendTo: listener_ip : (9999 + i)
   ↓
   ↓ real Linux UDP
```

```
   ↓
ROS2 Listener i (container)
UDP recv on port (9999 + i)
```

This creates a full round-trip path:
real ROS2 → simulated wireless channel → real ROS2,
allowing controlled testing of latency, throughput, and packet loss under configurable wireless conditions.

2. ns-3 Wi-Fi 7 Simulation Study / Validation

Before coupling ROS2 with ns-3, I performed a separate set of simulations to check whether **ns-3's 802.11be model behaves realistically** under different traffic and PHY configurations.

Parameters explored:

- **Packet sizes:** 0.1 KB → 0.5 KB (small), 4–16 KB (medium), 64 KB (large)
- **Channel widths:** 80 MHz / 160 MHz
- **Bands:** 5 GHz / 6 GHz
- **Traffic load:** multi-STA uplink with controlled Mbps per STA
- **Metrics:** latency, throughput, drop rate

Key observations:

- Small packets exhibit higher relative overhead (higher latency, lower throughput).
- Medium packets achieve highest MAC-layer efficiency ("MAC sweet spot").
- Large packets increase latency and jitter due to increased airtime and higher loss probability.
- 160 MHz channels show better throughput under load, as expected.
- 6 GHz generally has slightly lower latency due to cleaner spectrum.

These trends match expected Wi-Fi 6/7 behavior, suggesting the ns-3 model is producing reasonable results and is suitable for further thesis work.

3. Current Status

- The ROS2 ↔ ns-3 integration works end-to-end.
- Wireless behavior has been preliminarily validated.
- The final thesis research question is **still open**.

4. What I am Looking For

I would appreciate advice on:

- whether this direction could evolve into a suitable Master thesis topic, and
- whether there are groups more closely aligned with **networking, wireless systems, robotics networking, or cyber-physical communication** that might be a good fit.